Logistic Regression - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Step 1: Importing Libraries

 # Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    roc_auc_score, roc_curve, confusion_matrix, classification_report
)
from ipywidgets import interact, Dropdown
print("Libraries Imported");
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

 # Read and store the Dengue dataset

```
data = pd.read_csv('Dengue_dataset.csv')
print("Dataset reading complete")
```

Output:

```
Dataset reading complete
```

Step 3: Data Analysis

 # Display the first 5 rows of the dataset

```
data.head()
```

Output:

```
 index  Age  Fever Days  Platelets  Hematocrit   WBC   Headache  EyePain  Muscle Pain  Joint
 Pain  Rash  Nausea  Vomiting  Abdominal Pain  Bleeding  Lethargy  Dengue
 0      61   6           56777      49.49572857  3205  0         1        0            0
 0     1       1         0              0         0         1
 1      24   7           61250      42.41093608  3738  1         0        0            0
 1     0       0         1              1         0         1
 2      70   7           88034      49.66143051  3052  1         0        1            0
 0     1       0         0              0         0         1
 3      30   6           55130      50.20159282  2713  1         0        0            1
 0     0       1         1              0         0         1
 4      33   3           64346      43.46980401  4888  1         1        0            1
 1     1       1         0              1         0         1
```

Display the last 5 rows of the dataset

```
data.tail()
```

Output:

```
 index  Age  Fever Days  Platelets  Hematocrit   WBC   Headache  EyePain  Muscle Pain  Joint
 Pain  Rash  Nausea  Vomiting  Abdominal Pain  Bleeding  Lethargy  Dengue
 4995   59   2           123791     47.18861033  4239  1         0        0            0
 0     0       0         0              0         0         0
 4996   11   4           111575     41.84560446  5989  1         0        1            1
 1     1       1         0              0         0         0
 4997   19   2           128198     40.2842782   4775  1         1        1            0
 0     0       0         0              0         1         0
 4998   11   0           141619     47.3939162   4625  1         1        0            0
 1     1       1         1              0         1         0
 4999   69   2           90619      46.47148497  5982  1         0        0            0
 1     0       1         0              0         0         0
```

Show statistical summary of numerical columns

```
data.describe()
```

Output:

```
 Age     Fever Days  Platelets  Hematocrit  WBC      Headache  Eye Pain  Muscle Pain  Joint Pain
 Rash    Nausea  Vomiting  Abdominal Pain  Bleeding  Lethargy  Dengue
 count  5000.0  5000.0      5000.0     5000.0      5000.0   5000.0    5000.0    5000.0
 5000.0      5000.0  5000.0  5000.0    5000.0          5000.0    5000.0    5000.0
 mean   41.85   3.41        141522.48  41.86       6156.58  0.517     0.492     0.508
 0.500       0.496   0.501   0.507     0.496           0.226     0.502     0.460
 std    18.80   2.05        87116.73   4.81        2594.06  0.500     0.500     0.500
 0.500       0.500   0.500   0.500     0.500           0.418     0.500     0.498
 min    10.0    0.0         15060.0    32.02       2002.0   0.0       0.0       0.0          0.0
 0.0     0.0     0.0       0.0             0.0       0.0       0.0
 25%    26.0    2.0         62365.5    38.43       3848.75  0.0       0.0       0.0          0.0
 0.0     0.0     0.0       0.0             0.0       0.0       0.0
 50%    42.0    3.0         130972.0   42.02       6169.0   1.0       0.0       1.0          0.5
 0.0     1.0     1.0       0.0             0.0       1.0       0.0
 75%    58.0    5.0         221014.25  44.89       8302.5   1.0       1.0       1.0          1.0
 1.0     1.0     1.0       1.0             0.0       1.0       1.0
 max    74.0    7.0         299988.0   51.99       10993.0  1.0       1.0       1.0          1.0
 1.0     1.0     1.0       1.0             1.0       1.0       1.0
```

Display dataset structure and data types

```
data.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 16 columns):
#   Column         Non-Null Count  Dtype
0   Age            5000 non-null   int64
1   FeverDays      5000 non-null   int64
2   Platelets      5000 non-null   int64
3   Hematocrit     5000 non-null   float64
4   WBC            5000 non-null   int64
5   Headache       5000 non-null   int64
6   EyePain        5000 non-null   int64
7   MusclePain     5000 non-null   int64
8   JointPain      5000 non-null   int64
9   Rash           5000 non-null   int64
10  Nausea         5000 non-null   int64
11  Vomiting       5000 non-null   int64
12  AbdominalPain  5000 non-null   int64
13  Bleeding       5000 non-null   int64
14  Lethargy       5000 non-null   int64
15  Dengue         5000 non-null   int64
dtypes: float64(1), int64(15)
memory usage: 625.1 KB
```

Check the number of missing values in each column

```
data.isnull().sum()
```

Output:

```
Column         Missing Values
Age            0
FeverDays      0
Platelets      0
Hematocrit     0
WBC            0
Headache       0
EyePain        0
MusclePain     0
JointPain      0
Rash           0
Nausea         0
Vomiting       0
AbdominalPain  0
Bleeding       0
Lethargy       0
Dengue         0
dtype: int64
```

Show the number of rows and columns in the dataset

```
data.shape
```

Output:

```
(5000, 16)
```

Count occurrences of each class in the Dengue column

```
data['Dengue'].value_counts()
```

Output:
```
0    27001    2300Name: Dengue, dtype: int64
```

 # Count frequency of each Platelets value

```
data['Platelets'].value_counts()
```

Output:
```
Platelets
244845    2
32653     2
26618     2
61015     2
58169     2
...
25367     1
72241     1
18385     1
51701     1
70182     1
4961 rows x 1 columns
dtype: int64
```

 # Count frequency of each WBC value

```
data['WBC'].value_counts()
```

Output:
```
WBC
6916     6
6573     5
9771     5
2521     5
7405     5
...
3987     1
4247     1
5815     1
3598     1
5860     1
3781 rows x 1 columns
dtype: int64
```

Step 4: Data Preprocessing

 # Encode Dengue column by flipping values (0 -> 1, 1 -> 0)

```
data = data.replace({'Dengue':{0:1, 1:0}})
data.head()
```

Output:

```
Dataset Preview:
index  Age  FeverDays  Platelets  Hematocrit  WBC   Headache  EyePain  MusclePain  JointPain
Rash  Nausea  Vomiting  AbdominalPain  Bleeding  Lethargy  Dengue
0      61   6          56777      49.4957     3205  0         1        0           0          0
1       1         0             0         0         0
1      24   7          61250      42.4109     3738  1         0        0           0          1
0       0         1             1         0         0
2      70   7          88034      49.6614     3052  1         0        1           0          0
1       0         0             0         0         0
3      30   6          55130      50.2016     2713  1         0        0           1          0
0       1         1             0         0         0
4      33   3          64346      43.4698     4888  1         1        0           1          1
1       1         0             1         0         0
```

Encode Headache column by flipping values (0 -> 1, 1 -> 0)

```
data = data.replace({'Headache':{0:1, 1:0}})
data.head()
```

Output:

```
Dataset Preview:
index  Age  FeverDays  Platelets  Hematocrit  WBC   Headache  EyePain  MusclePain  JointPain
Rash  Nausea  Vomiting  AbdominalPain  Bleeding  Lethargy  Dengue
0      61   6          56777      49.4957     3205  1         1        0           0          0
1       1         0             0         0         0
1      24   7          61250      42.4109     3738  0         0        0           0          1
0       0         1             1         0         0
2      70   7          88034      49.6614     3052  0         0        1           0          0
1       0         0             0         0         0
3      30   6          55130      50.2016     2713  0         0        0           1          0
0       1         1             0         0         0
4      33   3          64346      43.4698     4888  0         1        0           1          1
1       1         0             1         0         0
```

Encode Rash column by flipping values (0 -> 1, 1 -> 0) and verify changes

```
data = data.replace({'Rash':{0:1, 1:0}})
print("Encoding complete. Checked head:")
data.head()
```

Output:

```
Encoding complete. Checked head:
Dataset Preview:
index  Age  FeverDays  Platelets  Hematocrit  WBC   Headache  EyePain  MusclePain  JointPain
Rash  Nausea  Vomiting  AbdominalPain  Bleeding  Lethargy  Dengue
0      61   6          56777      49.4957     3205  1         1        0           0          1
1       1         0             0         0         0
1      24   7          61250      42.4109     3738  0         0        0           0          0
0       0         1             1         0         0
2      70   7          88034      49.6614     3052  0         0        1           0          1
1       0         0             0         0         0
3      30   6          55130      50.2016     2713  0         0        0           1          1
0       1         1             0         0         0
4      33   3          64346      43.4698     4888  0         1        0           1          0
1       1         0             1         0         0
```

Step 5: Model Training

 # Set the target variable, separate features and labels, and split the dataset into 80% training and 20% testing data

```
target_col = "Dengue"
print("Target column set to:", target_col)
X = data.drop(columns=['Platelets', target_col])
y = data[target_col]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42,
    stratify=y)
print(f"Split: 80% Train ({len(X_train)}) / 20% Test ({len(X_test)}) samples")
```

Output:

```
Target column set to: Dengue
Split: 80% Train (4000) / 20% Test (1000) samples
```

 # Display the feature matrix

```
X
```

Output:

```
Feature Matrix (X) Preview:
Age  FeverDays  Platelets  Hematocrit  WBC   Headache  EyePain  MusclePain  JointPain  Rash
Nausea  Vomiting  AbdominalPain  Bleeding  Lethargy
0     61   6          56777      49.495729   3205  0         1        0           0          0
1       1         0              0          0
1     24   7          61250      42.410936   3738  1         0        0           0          1
0       0         1              1          0
2     70   7          88034      49.661431   3052  1         0        1           0          0
1       0         0              0          0
3     30   6          55130      50.201593   2713  1         0        0           1          0
0       1         1              0          0
4     33   3          64346      43.469804   4888  1         1        0           1          1
1       1         0              1          0
...   ...  ...        ...        ...          ...   ...       ...      ...         ...
...   ...     ...       ...            ...       ...
4995  59   2          123791     47.188610   4239  1         0        0           0          0
0       0         0              0          0
4996  11   4          111575     41.845604   5989  1         0        1           1          1
1       1         0              0          0
4997  19   2          128198     40.284278   4775  1         1        1           0          0
0       0         0              0          1
4998  11   0          141619     47.393916   4625  1         1        0           0          1
1       1         1              0          1
4999  69   2          90619      46.471485   5982  1         0        0           0          1
0       1         0              0          0
5000 rows x 15 columns
```

 # Display the first few values of the target variable

```
y.head()
```

Output:

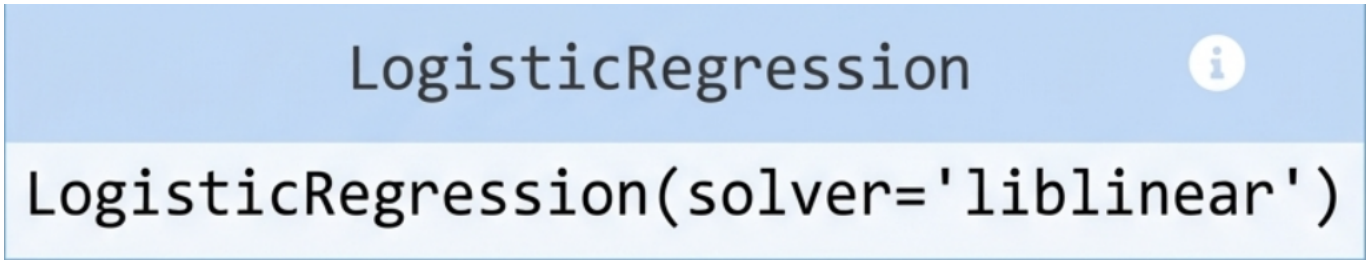
```
Target Vector (y) Preview:
Dengue
0  0
1  0
2  0
3  0
4  0
Name: Dengue, dtype: int64
```

 # Initialize and train a Logistic Regression model with L2 regularization on the training data

```
model = LogisticRegression(C=1.0, penalty='l2', solver='liblinear')
model.fit(X_train, Y_train)
print("Model trained successfully")
```

Output:

```
Model trained successfully
```



Step 6: Model Evaluation

 # Preparing Feature and Target Data for Logistic Regression Analysis

```
numeric_cols = X.select_dtypes(include=[np.number]).columns.tolist()
feature_names = numeric_cols
X_num = X[feature_names]
y_arr = np.array(y)
print(f"Feature-target data ready -> {len(feature_names)} numeric features selected for
    modeling")
```

Output:

```
Feature-target data ready -> 14 numeric features selected for modeling
```

 # Fits a logistic regression model on a single feature and visualizes the resulting sigmoid probability curve against th

```
def show_1d_sigmoid(f):
    X,y = X_num[[f]].values, y_arr; c = LogisticRegression(max_iter=5000).fit(X,y)
    a,b = X.min(),X.max();
    a,b = (a-1,b+1) if a==b else (a,b); g= np.linspace(a,b,300)[:,None]; p =
    c.predict_proba(g)[:,1]
    plt.scatter(X[y==0],y[y==0]); plt.scatter(X[y==1],y[y==1]); plt.plot(g,p);
    plt.xlabel(f);
    plt.ylabel("Probability of Dengue");
    plt.ylim(-.1,1.1); plt.grid(1); plt.title(f"Sigmoid using {f}");
    plt.show()

print("Function defined: show_1d_sigmoid")
```

Output:

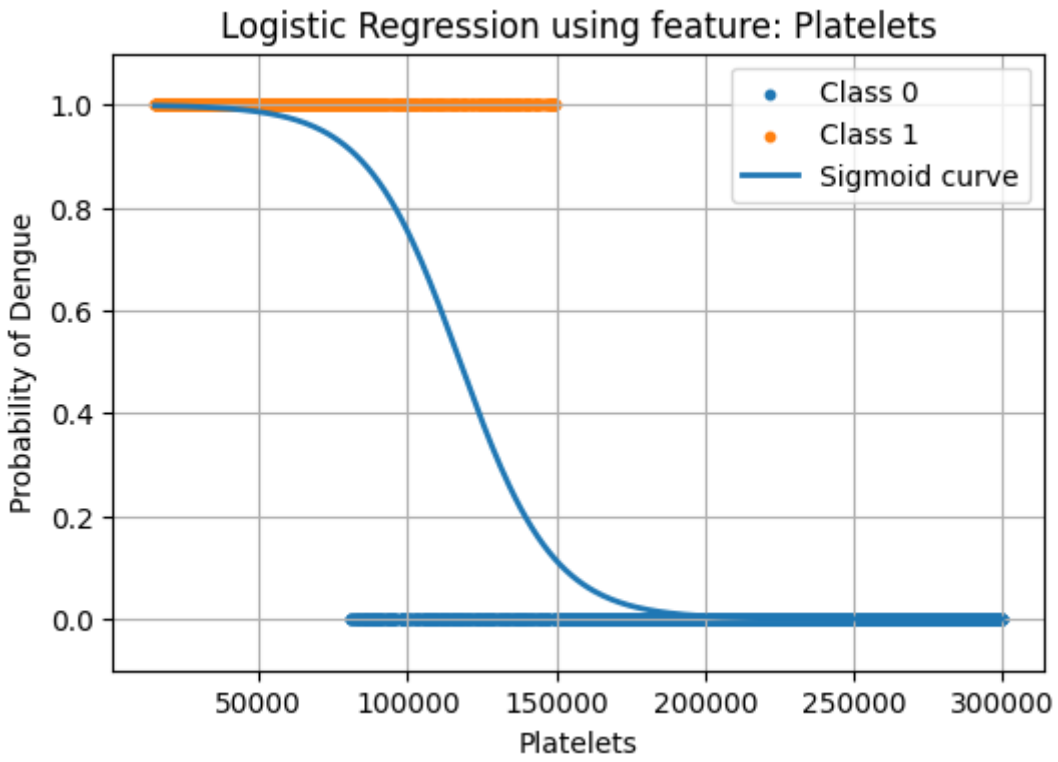
```
Function defined: show_1d_sigmoid
```

Interactive dropdown over all features

```
interact(
    show_1d_sigmoid,
    feat=Dropdown(options=feature_names, description="Feature")
)
```

Output:

```
Feature
Age
FeverDays
Platelets
Hematocrit
WBC
MusclePain
JointPain
Nausea
Vomiting
AbdominalPain
Bleeding
Lethargy
Rash
EyePain
Headache
```



Training Performance

```
Y_train_pred = model.predict(X_train)
print(" Training Performance ")
print(f"Accuracy : {accuracy_score(y_train, Y_train_pred):.4f}")
print(f"Precision: {precision_score(y_train, Y_train_pred):.4f}")
print(f"Recall   : {recall_score(y_train, Y_train_pred):.4f}")
print(f"F1 Score : {f1_score(y_train, Y_train_pred):.4f}")
```

Output:

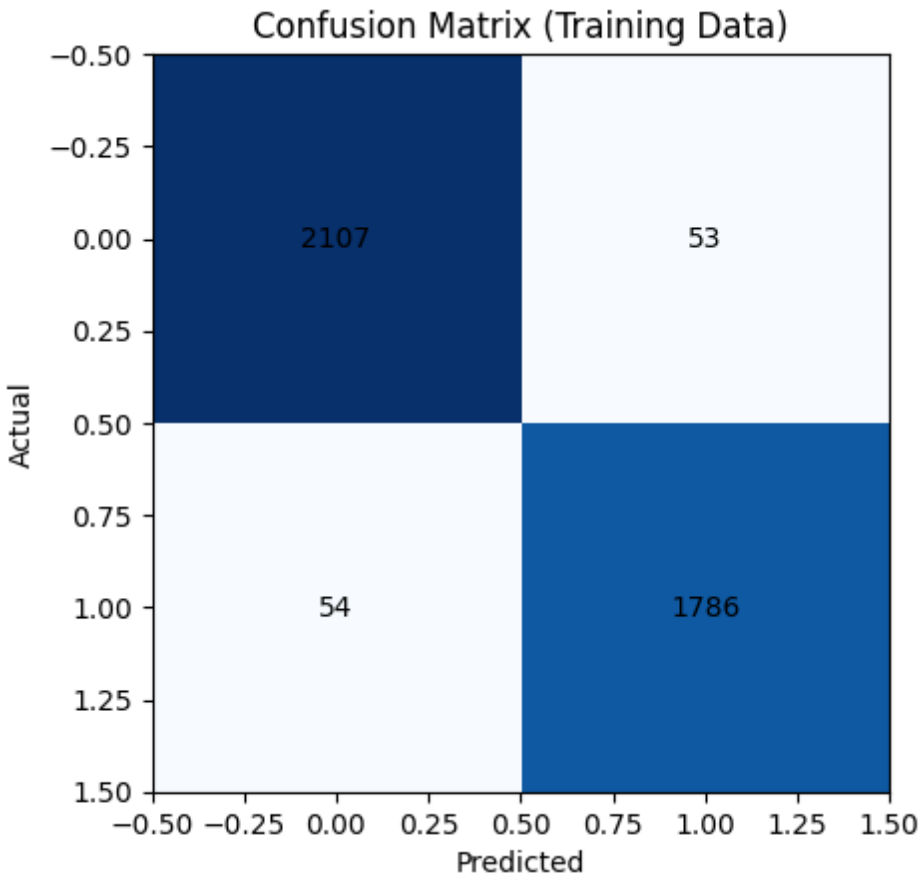
```
Training Performance
Accuracy : 0.9732
Precision: 0.9712
Recall   : 0.9707
F1 Score : 0.9709
```

Confusion Matrix for training data

```
cm_train = confusion_matrix(y_train, Y_train_pred)
sns.heatmap(cm_train, cmap="Blues")
plt.title("Confusion Matrix (Training Data)")
for i in range(len(cm_train)):
    for j in range(len(cm_train)):
        plt.text(j, i, cm_train[i][j], ha='center', va='center')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

Output:

```
Confusion Matrix (Training Data)
```



Testing Performance

```
Y_test_pred = model.predict(X_test)
print("Testing Performance")
print(f"Accuracy : {accuracy_score(y_test, Y_test_pred):.4f}")
print(f"Precision: {precision_score(y_test, Y_test_pred):.4f}")
print(f"Recall   : {recall_score(y_test, Y_test_pred):.4f}")
print(f"F1 Score : {f1_score(y_test, Y_test_pred):.4f}")
```

Output:

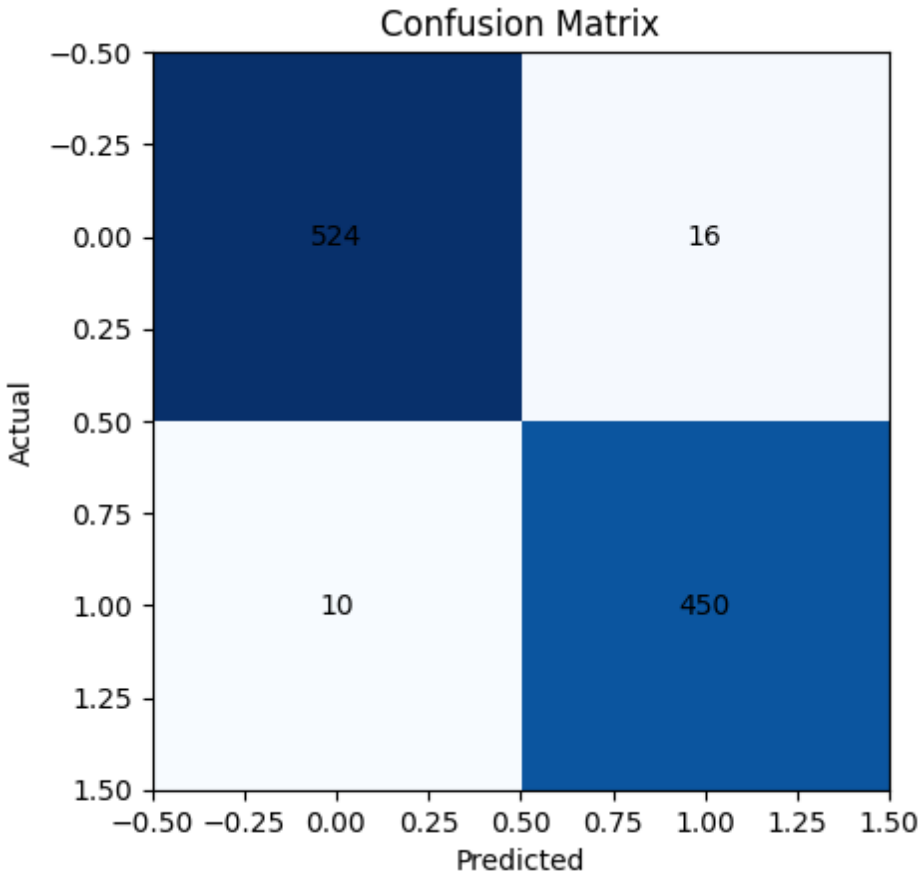
```
Testing Performance
Accuracy : 0.9740
Precision: 0.9657
Recall   : 0.9783
F1 Score : 0.9719
```

Confusion Matrix for testing data

```
cm_test = confusion_matrix(y_test, Y_test_pred)
sns.heatmap(cm_test, annot=True, cmap="Blues", fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix for Testing Data")
plt.show()
```

Output:

```
Confusion Matrix (Testing Data)
```



 # Calculates predicted probabilities to compute ROC curve metrics and the AUC score.

```
Y_proba = model.predict_proba(X_test)[:,1]
fpr, tpr, th = roc_curve(Y_test, Y_proba)
roc_auc = auc(fpr, tpr)
print(f"ROC metrics calculated. AUC = {roc_auc:.3f}")
```

Output:

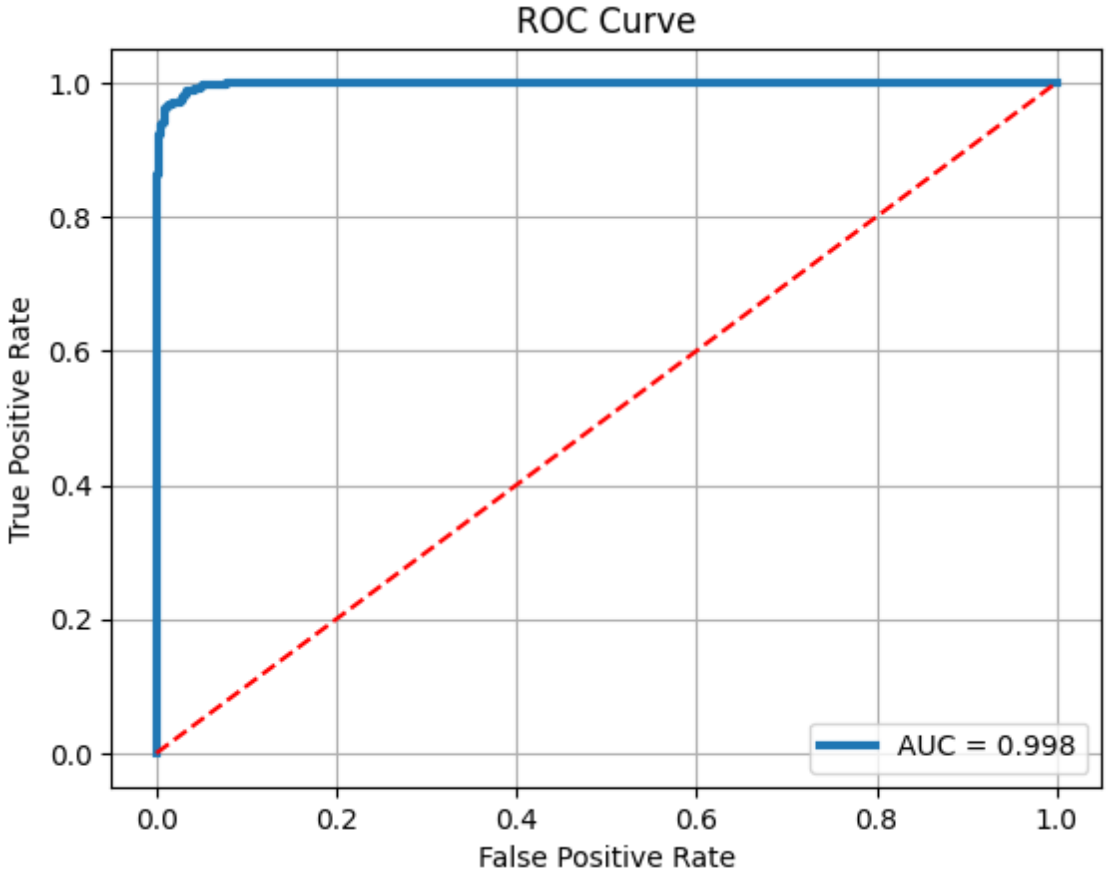
```
ROC metrics calculated. AUC = 0.998
```

 # Plots the ROC curve including the AUC score and a random classifier baseline.

```
plt.plot(fpr, tpr, linewidth=3, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], 'r--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.grid()
plt.show()
```

Output:

```
ROC Curve
```



 # Random Test Prediction

```
test_samples=data.sample(5)
display(test_samples)
sample = X_test.sample(1)
actual = Y_test.loc[sample.index].values[0]
pred = model.predict(sample)[0]
prob = model.predict_proba(sample)[0][1]
print("------ Random Sample Test ------")
print(sample)
print("Actual Dengue:", actual)
print("Predicted Dengue:", pred)
print("Probability:", prob)
```

Output:

```
Select a row to see prediction:
Index  Age  FeverDays  Hematocrit  WBC  Headache  EyePain  MusclePain  JointPain  Rash  Nausea
Vomiting  AbdominalPain  Bleeding  Lethargy
```